

# A Theory of Climate

## (or Theory in Climate?)

Is there an underlying theory of the 'whole thing', rather like in evolution?

No: In the sense that there are no easily stated grand principles that underpin the enterprise, aside from the basic laws of physics, chemistry etc.

But simpler theory is important:

- Improving GCMs. Tomorrow's generation of numerical models will depend on today's theory.
- As an end in itself, as part of scientific progress.

Two particular classes of theory are needed:

- 1 Subgrid processes where models are imperfect.
- 2 The behavior of the system as a whole.

If there comes a time when theory is permanently stalled, then progress in the field halts and the subject stagnates. The end of that branch of science.